Rishiraj Nagarajan

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Education

Carnegie Mellon University — Pittsburgh, PA
Master of Computational Data Science — *Expected Dec 2026*

Relevant Coursework: Large Language Model Systems; Introduction to Machine Learning; LLMs and Their Applications; AI Agents; Introduction to Deep Learning; Deep Reinforcement Learning and Control

Vellore Institute of Technology — Vellore, India
B.Tech. in Computer Science Engineering — *Jun 2025*

Experience

Machine Learning Engineer Intern | Different Labs (different.ai) — May 2026 – Aug 2026
Backed by General Catalyst | Python, Java, Gemini Vision, Temporal, MongoDB, PostgreSQL, Redis

- Built shared preference memory for Java and Python content pipelines serving 10+ organizations; removed stale, duplicate, or conflicting preferences before generation to prevent repeat client inputs, saving 300 user-hours monthly.
- Built Gemini Vision profiles from client posts to condition generated content on color, imagery, layout, and text; in LLM-as-a-judge evaluation, scored 47% higher in brand alignment than the legacy direct-LLM web-check pipeline.
- Reduced duplicate workflow runs 25% with unique Temporal IDs, MongoDB execution checks, and idempotent retries; grounded editorial checks in source documents and added human review and rollback controls.

Graduate Researcher | Carnegie Mellon University LTI — Apr 2026 – Present
Advisor: Prof. Maarten Sap | Paper under review at EMNLP 2026

- Built SocialDrift, a 100-case benchmark measuring conversational drift in ambiguous social conflicts; across five-turn dialogues, LLMs' "you were right" judgments rose from 1% to 24%.
- Analyzed 36K judgments to isolate drift drivers: conversation context accounted for 64%, with self-justification the strongest trigger; evaluation on 100 Reddit AITA posts reached 73% agreement with crowd labels.

Teaching Assistant | Carnegie Mellon University — Aug 2026 – Present
11-868 Large Language Model Systems | CUDA, GPU acceleration, LLM systems

- Taught 120 students CUDA, GPU acceleration, speculative decoding, transformer inference, and TPU architectures.

Data Science Intern | Brandidea Granular Analytics — Jan 2025 – May 2025
Python, XGBoost, AWS Bedrock, Lambda, RAG, time-series forecasting

- Engineered seasonality-aware XGBoost features to forecast cement demand by district, reducing RMSE 15% versus an analyst forecast baseline.
- Built a RAG pipeline translating natural-language questions into validated district queries with age, gender, income, and education filters; achieved 92% parsing accuracy across 150 queries with sub-2s latency.

Projects

Structure-Aware KV Cache Eviction | SGLang, PyTorch — Jan 2026 – Present
Upstream SGLang pull request under review

- Implemented weighted-LFU inference optimization in SGLang's radix-tree KV cache, weighting reuse by prefix depth; on MS MARCO prompt-reuse workloads, cut P99 time to first token 39.6% and mean latency 26% versus LRU.

APIShift: Tool-Using LLM Adaptation — Jul 2026
GRPO, LoRA, vLLM, Modal

- Built APIShift, an evaluation benchmark and agent-orchestration pipeline testing whether LLM agents recover from silent API changes by inspecting errors or live docs and adapting tool calls; success is verified from final API state.
- Across 3 seeds, reinforcement-learning fine-tuning with GRPO raised Qwen3-4B seen-change success from 77% to 85%, but held-out success was 38% vs. 40% before training (no transfer); a recovery prompt raised it from 40% to 52%.

Validator for LLM-Generated Charts | CMU AI Agents — Sep 2026
Python, Matplotlib, Modal, pytest

- Built automated evals that replay chart-agent commands in a sandbox and inspect instrumented Matplotlib output for execution, data, design, and readability errors, checking axes, labels, plotted values, text overlap, and clipping; raised macro-MCC from 0.03 for a vision-language judge baseline to 0.67 across 110 labeled runs.

Technical Skills

Languages: Python, Go, Java, TypeScript, C++, SQL, CUDA
ML and LLM Systems: PyTorch, XGBoost, SGLang, vLLM, GRPO, LoRA, RAG, LangChain
Cloud and Data: AWS (EKS, Lambda, RDS, Bedrock), Modal, Docker, Kubernetes, Terraform, Temporal, MongoDB, PostgreSQL, Redis, FastAPI